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Topic: Heart Attack Analysis & Prediction

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ABSTRACT

This report represents the major project assigned to the participants of the Summer Internship program Conducted by the Silicon Institute of Technology Bhubaneswar on the field of Machine Learning using Python. Cardiovascular diseases are the most common day diseases among the population of the globe. This is believed to be the most common cause of the death in the few decades in the developed as well as the developing countries. Early detection of the cardiac diseases and continuous supervision of the clinicians can reduce the mortality rate. However, it is not possible to monitor the patients for 24x7 by a doctor since it requires more sapience. In this project I have developed and researched about models for heart disease prediction through the various heart attributes of a patient and detect impending heart disease using Machine Learning techniques like Logistic Regression, KNN Classification algorithm on the dataset available publicly in Kaggle Website, further evaluating the results using the confusion Matrix and classification report. The early diagnosis of the diseases can lead to a decrease in the death rate and high risks and hence reduce the complications.

Key Words: Machi8ne Learning, Logistic Regression, KNN classification Regression, Confusion Matrix, Cardiovascular Diseases.

- **Introduction:**

According to the World Health Organization, every year 12 million deaths occur worldwide due to Heart Disease. The load of cardiovascular disease is rapidly increasing all over the world from the past few years. Many researches have been conducted in attempt to pinpoint the most influential factors of heart disease as well as accurately predict the overall risk. Heart Disease is even highlighted as a silent killer which leads to the death of the person without obvious symptoms. The early diagnosis of heart disease plays a vital role in making decisions on lifestyle changes in high-risk patients and in turn reduce the complications. This project aims to predict future Heart Disease by analyzing data of patients which classifies whether they have heart disease or not using machine-learning algorithms.

- **Problem Definition:**

The major challenge in this era is of predicting the chance of heart rate if a person has or not using some given attributes about the disease. There are many wonders of Technology that helps in this field but is either too expensive or unachievable by a normal citizen. The abundant data present in the world regarding the same topic can help in the formulation of the model using the Machine Learning Mechanism to predict the outcome of the disease using some given number of inputs. The hidden patterns and relations can be used for the health diagnosis and medical data.

Objective:

The main objective of the project is to predict the future possibility of heart disease by implementing suitable machine Learning Algorithm.

- **Dataset:**

I have used the **"heart.csv"** dataset from Kaggle platform which is a comma separated values (CSV) files with 303 rows and 14 attributes. The data represents the survey of as many people in the dataset about the Cardiovascular Diseases. Pandas library of the python has been used to read the values of the dataset and further converted to a NumPy array with distinct values.

	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	0
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	0
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	0
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	0
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	0

303 rows × 14 columns

Fig. 1: Snapshot of Actual dataset (heart.csv)

As shown in the figure above the attributes of the dataset are depicting the values as follows:

i. Age: Age of the person
ii. Sex: gender (Male : 1/ Female : 0)
iii. cp : Chest pain type
iv. trtbps: resting blood pressure (in mm Hg)
v. chol: cholesterol in mg/dl
vi. fbs: fasting blood sugar
vii. restecg: resting electrocardiographic results
viii. thalachh: maximum heart rate
ix. exng: exercise induced angina (1: yes, 0: No)
x. oldpeak
xi. slp: Slope
xii. caa; no. of major vessels
xiii. thall: Thal rate
xiv. Output: Target Variable

The dataset contains one duplicate value and 0 NULL values. The duplicate row has been dropped in order to avoid any casualty during the further process. The unnecessary columns i.e. "slp", "thall" and "caa" had also been dropped. The new dataset now remained with 302 rows and 11 columns.

- **Algorithms Used:**

In order to predict I have the ML model using KNN (K Nearest Neighbours) Algorithm and Logistic Regression algorithm. The data set is a classification type due to which these algorithms best suited.

i. Logistic Regression Algorithm:

Logistic Regression is a supervised classification algorithm. It is a predictive analysis algorithm based on the concept of probability. It measures the relationship between the dependent variable (TenyearCHD) and the one or more independent variables (risk factors) by estimating probabilities using underlying logistic function (sigmoid function). Sigmoid function is used as a cost function to limit the hypothesis of logistic regression between 0 and 1 (squashing) i.e. $0 \leq h\theta(x) \leq 1$.

In logistic regression cost function is defined as:

$$Cost(h\theta(x), y) = \begin{cases} -\log\big(h\theta(x)\big), & if\ y = 1 \\ -\log\big(1 - h\theta(x)\big), & if\ y = 0 \end{cases}$$

Logistic Regression relies highly on the proper presentation of data. So, to make the model more powerful, important features from the available data set are selected using backward elimination and recursive elimination techniques.

ii. KNN (K Nearest Neighbours Algorithm):

KNN Algorithm is one of the simplest machine Learning algorithm based on supervised Learning technique. The KNN algorithm assumes the similarity between the new case data and put the new case into the category that is most similar to the available categories. It stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. It can be used for Regression as well as for classification but mostly it is used for the classification problems. It is

used to be known as the non-parametric algorithm as it does not make any assumptions on the underlying data. KNN works on the principle of calculating and comparing the Euclidean distance of K number of neighbours. Then by counting the number of data points in each category among the k neighbours.

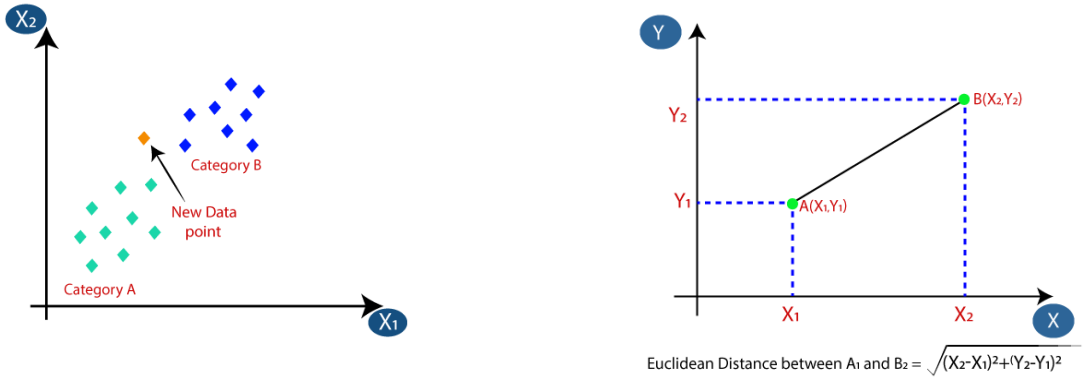


Fig 2: K- nearest Neighbour Algorithm

- **Discussion on results:**

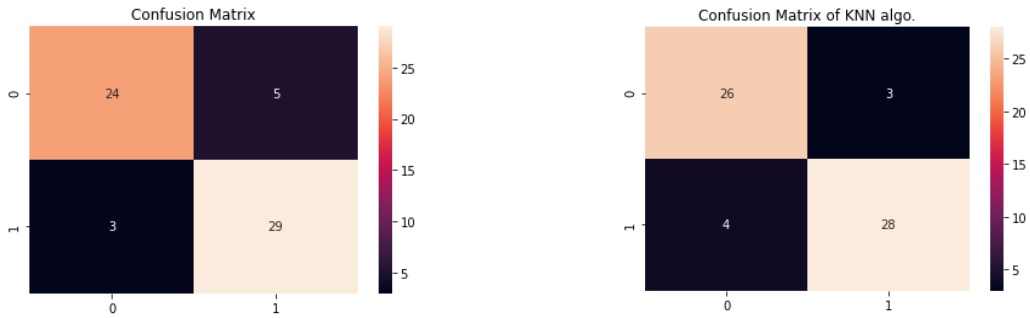
The model was built and the training data was passed into the model to get the predicted output using both KNN as well as Logistic Classification. And it was compared with the corresponding test data of the feature Matrix. The Performance metrics had been calculated for both the algorithms. And the following data has been observed.

Algorithms	Accuracy	Precision
Logistic Regression	86.88%	85.29%
KNN (k = 5)	88.5%	90.32%

From the above table it has been seen that the KNN Algorithm has more accurate classification rather than the Logistic regression one.

The confusion Matrix:

The fundamental of a confusion matrix is the number of correct and incorrect predictions which are sum of classes. The numbers represented in the lighter colour in the heat Map is the value that are correctly mapped to the test data and those values are mismatched they are shown in black boxes.



(a) *(b)*

Fig 3: The heat map visualization of Confusion Matrices for (a) Logistic & (b) KNN Algo.

Libraries Used:

i. numpy
ii. pandas
iii. Scikit-learn(sklearn)
iv. Matplotlib
v. Seaborn

Code:

The link to the code i.e. the IPYNB notebook file has been provided below for the visualization and references.

https://github.com/swayansu/Machine_Learning-/tree/main/Major%20project

- **CONCLUSION:**

The early prognosis of cardiovascular diseases can aid in making decisions on lifestyle changes in high risk patients and in turn reduce the complications, which can be a great milestone in the field of medicine. This project classified and predicted the outputs with an accuracy of ~89% and precision of 90%. Further for its enhancement, we can train on models and predict the types of cardiovascular diseases providing recommendations to the users, and also use more enhanced models.

- **REFERENCES:**

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